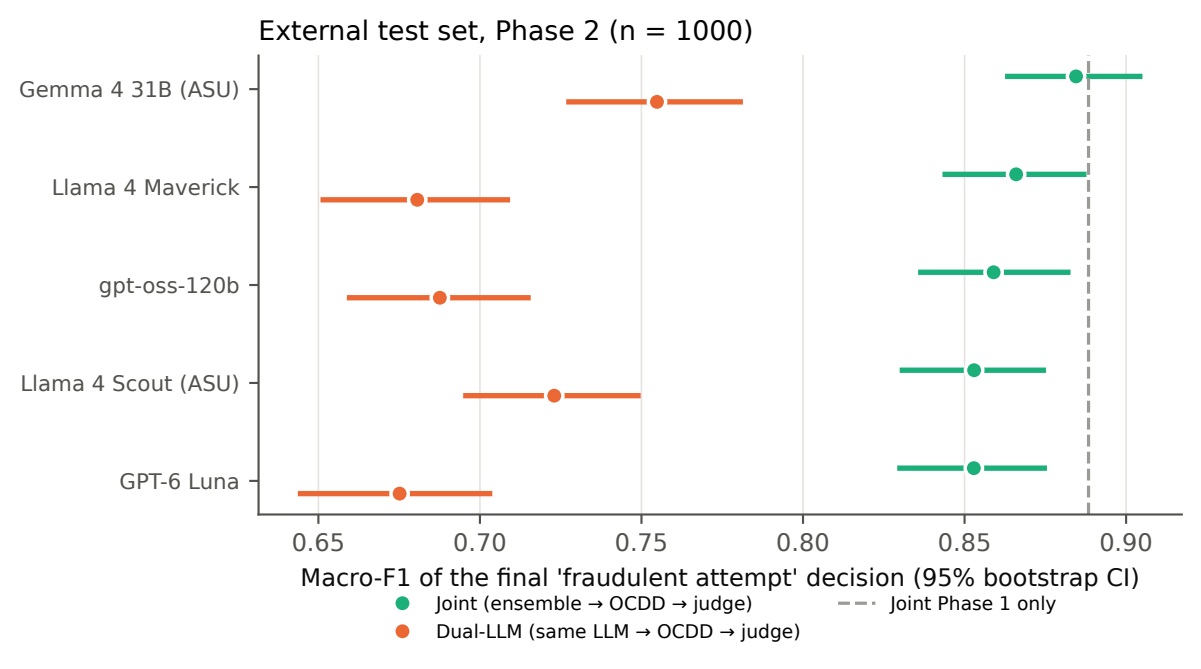
External test set, Phase 2 (n = 1000)
Gemma 4 31B (ASU)
Llama 4 Maverick
gpt-oss-120b
Llama 4 Scout (ASU)
GPT-6 Luna
0.65
0.70
0.75
0.80
0.85
0.90
Macro-F1 of the final 'fraudulent attempt' decision (95% bootstrap CI)
Joint (ensemble → OCDD → judge)
Dual-LLM (same LLM → OCDD → judge)
Joint Phase 1 only